Steps to Configure AWS Linux Host for Ansible

1. Launch the EC2 Instance

- Go to AWS Console → EC2 → Launch Instance.

- Choose **Amazon Linux 2** or **Ubuntu** (both are common for Ansible labs).

- Select instance type (e.g., `t2.micro` for testing).

- Configure networking (ensure **SSH (port 22)** is open in the Security Group).

- Attach a key pair for SSH login.

2. Connect to the Instance

```bash
ssh -i ~/.ssh/mykey.pem ec2-user@<EC2_PUBLIC_IP>
```

- Default user:

  - Amazon Linux → `ec2-user`

  - Ubuntu → `ubuntu`

3. Prepare the Host for Ansible

- Update packages:

  ```bash
  sudo yum update -y   # Amazon Linux
  sudo apt update -y   # Ubuntu
  ```

- Install Python (needed for Ansible modules):

  ```bash
  sudo yum install -y python3   # Amazon Linux
  sudo apt install -y python3   # Ubuntu
  ```

- Verify:

  ```bash
  python3 --version
  ```

4. Create a Dedicated Ansible User (Optional but Recommended)

```bash
sudo adduser ansibleu
sudo passwd ansibleu
sudo usermod -aG wheel ansibleu   # Amazon Linux
sudo usermod -aG sudo ansible    # Ubuntu
```

- Configure **SSH key-based login**:

  ```bash
  sudo mkdir /home/ansibleu/.ssh
  sudo chown -R ansibleu:ansibleu /home/ansibleu/.ssh
  sudo chmod 700 /home/ansibleu/.ssh
  sudo touch authorized_keys
  sudo chmod 600 /home/ansibleu/.ssh/authorized_keys
  ```

- on your control Node create **SSH key**:

```
ssh-keygen -t rsa -b 4096 -f ~/.ssh/mykey
```

This creates:

- (private key)
- (public key)

- copy `mykey.pub` to `authorized_keys` in `/home/ansibleu/.ssh/authorized_keys` in your EC2-instance.

5. Enable Passwordless Sudo

Edit sudoers:

```bash

sudo visudo

```

Add:

```

ansibleu ALL=(ALL) NOPASSWD:ALL

```

6. Configure Ansible Control Node

On your **Ansible control machine** (could be your laptop or another server):

- Create inventory file:

  ```ini

  [aws_hosts]

  aws1 ansible_host=<EC2_PUBLIC_IP> ansible_user=ansibleu
ansible_ssh_private_key_file=~/.ssh/mykey

  ```

- Test connectivity:

  ```bash

  ansible aws_hosts -m ping

  ```

Expected output:

```json

aws1 | SUCCESS => {

  "ping": "pong"
```

TEST

- Create a file in current directory at yoour control node and run:

```bash
  ansible aws_hosts -m copy -a "src=test.txt dest=/home/ansible/"
```

Key Notes for Reliability:

- Always ensure **Security Group** allows SSH from your control node.

- Use **IAM roles** if you plan to integrate AWS modules (e.g., provisioning via Ansible).

- For reproducibility, script the setup with **cloud-init** or **Ansible playbooks** so every new EC2 host is ready automatically.

Thanks:

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